

SE Lab-1

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Problem Statement 56: Peer Skill Exchange & Mentorship Network

Complete Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-001	Functional	The system shall maintain a time-credit balance for each member, crediting 1 hour after a verified teaching session.	High	PASS: Mentor gains exactly 1 credit after a verified 1-hour session. FAIL: Credit is not added or is incorrect.	Core time-banking functionality.

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-002	Functional	The system shall allow a learner to search for and select a mentor based on skill and availability.	Medium	PASS: Matching mentors and available slots are displayed for a valid skill search. FAIL: Matching available mentors are not displayed.	Helps learners find suitable mentors.
FR-003	Functional	The system shall allow a learner to schedule a skill-exchange session with a selected mentor using an available time slot.	High	PASS: Booking is created when the slot is available and sufficient credits exist. FAIL: Booking is created when either condition is false.	Required to organize skill-exchange sessions.

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-004	Functional	The system shall verify completion of a teaching session through confirmation from both the learner and mentor before transferring time credits.	High	PASS: Credits transfer only after both participants confirm completion. FAIL: Credits transfer without both confirmations.	Prevents incorrect or fraudulent credit transfers.
FR-005	Functional	The system shall allow members to provide a rating and feedback after a completed skill-exchange session.	Low	PASS: Member can submit a rating from 1–5 after completion. FAIL: Rating is outside 1–5 or feedback is allowed before completion.	Helps maintain trust and service quality.

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
NFR-001	Nonfunctional (Performance And Security)	The session scheduling calendar shall sync with Google Calendar and Outlook through iCal export feeds.	Medium	PASS: Exported session appears with correct date, time and details. FAIL: Event is missing or contains incorrect details.	Prevents scheduling conflicts.

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
NFR-002	Nonfunctional (Performance)	The system shall display mentor and session search results within 3 seconds under normal conditions.	Medium	PASS: At least 95 of 100 searches finish within 3 seconds. FAIL: Fewer than 95 searches meet the limit.	Ensures responsive searching.